

System Requirements Specification

SkillMatch AI: Student Project and Team Matching System

Course: CS425 Software Engineering

Assignment: Lab2: UseCase-driven Requirement Analysis

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Repository URL: <https://github.com/sam-adroit/SkillMatch>

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1. Introduction

1.1 Purpose

This System Requirements Specification document defines the functional and non-functional requirements for SkillMatch AI. The document applies the use-case driven requirements analysis approach from the course. It identifies system actors, major use cases, use-case diagrams, use-case descriptions, supplementary specifications, and glossary terms.

1.2 Scope

SkillMatch AI is a web based system that helps students create skill profiles, find project topics, apply to projects, and form teams. Admins or instructors can create project topics, review applications, manage teams, and monitor student participation. The system also includes AI based recommendations for project matching, teammate matching, and team skill gap warnings.

1.3 Project Repository

GitHub Repository URL: [Insert GitHub repository URL here]. This URL should point to the active project repository where the SRS document, vision document, source code, diagrams, and other project resources are committed and pushed regularly.

1.4 Definitions, Acronyms, and Abbreviations

AI means Artificial Intelligence. SRS means System Requirements Specification. CRUD means Create, Read, Update, and Delete. UC means Use Case. Admin refers to the instructor or system administrator who manages project topics, students, and teams.

1.5 References

Course materials: Lessons 1 to 5, Lab2 assignment instructions, and the sample Use Case Description document provided for the course. Project input: SkillMatch AI Vision Document.

2. Overall Description

2.1 Product Perspective

SkillMatch AI is an independent course project management application for one course or department. It includes a browser based user interface, backend services, a database, and an optional AI recommendation service. The AI service supports recommendations but does not replace instructor approval.

2.2 Product Functions

The system provides user login, student profile management, skill and interest management, project topic management, project browsing, project application submission, team creation, AI project recommendations, AI teammate matching, admin review, and team skill gap warnings.

2.3 User Classes and Characteristics

Students use the system to manage their profiles, browse projects, apply to projects, request AI recommendations, and form teams. Admins or instructors use the system to manage users, create project topics, review applications, manage teams, and view dashboards. The AI recommendation service acts as an external supporting system used by selected use cases.

2.4 Operating Environment

The system will run as a web application accessible from modern browsers. The backend may be implemented with Spring Boot or another course approved backend framework. The database may be relational. AI features may use an external AI API or local AI model depending on course constraints.

2.5 Assumptions and Dependencies

Users have internet access and a modern browser. Students provide enough profile information for useful recommendations. Admins create project topics before students apply. The AI feature depends on an available AI service. If the AI service is unavailable, normal browsing, application, and team management functions remain available.

3. Requirements Analysis Approach

This SRS organizes requirements around user goals and system behavior. Requirements are captured through use-case diagrams and use-case descriptions. The document separates functional requirements from supplementary non-functional requirements.

3.1 Problems, Needs, and Features Mapping

ID	Problem	Need	Feature
P1	Students have difficulty finding project topics that match their skills and interests.	Students need clear project search and personalized suggestions.	Browse projects, filter projects, request AI project recommendations.
P2	Teams may be formed without balanced skills.	Students and instructors need visibility into team skill coverage.	Team creation, teammate matching, skill gap warning.
P3	Project applications may be tracked through informal messages or spreadsheets.	Applications need to be stored and reviewed in one system.	Apply to project, review applications, application status.
P4	Admins need a simple way to publish project topics.	Admins need project topic management.	Create, update, view, and delete project topics.
P5	Students may not understand why a project is recommended.	Recommendations need short explanations.	AI generated match explanation.
P6	Instructors need to monitor participation and team formation.	Admins need dashboard views.	View admin dashboard and team status.

4. Use-Case Model

The use-case model contains the actors, use-case diagrams, and use-case descriptions for the major system goals.

4.1 Actors

Actor	Description
Student	A course participant who creates a profile, manages skills and interests, browses projects, applies to projects, requests recommendations, and forms teams.

Actor	Description
Admin / Instructor	A course instructor or system administrator who manages project topics, reviews applications, manages users, monitors teams, and views project status.
AI Recommendation Service	An external or internal AI service that receives project, student, and team data and returns recommendations, explanations, and skill gap warnings.

4.2 Use-Case Diagrams

The following diagrams show the main system boundary and actor goals for SkillMatch AI.

Figure 1. Overall use-case diagram for SkillMatch AI.

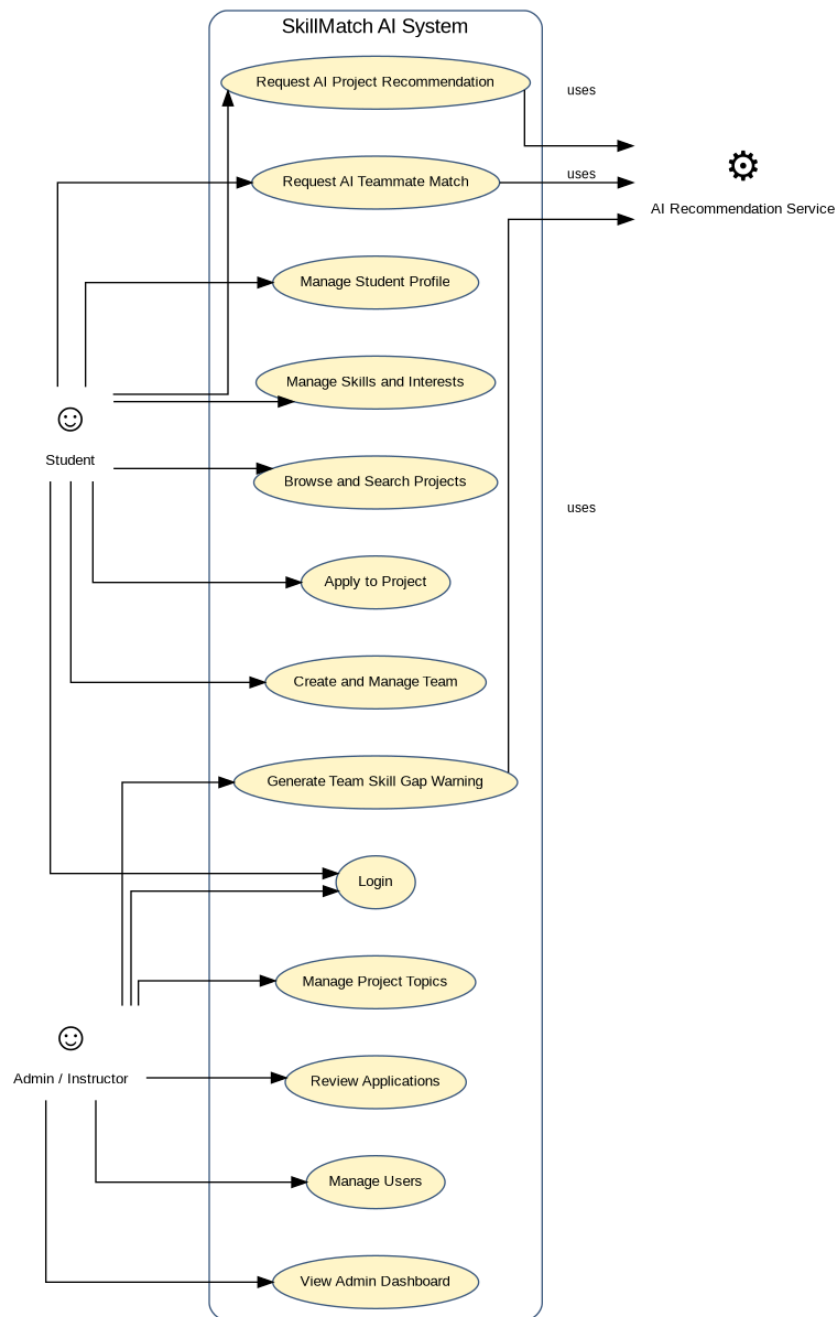


Figure 2. Student use-case diagram.

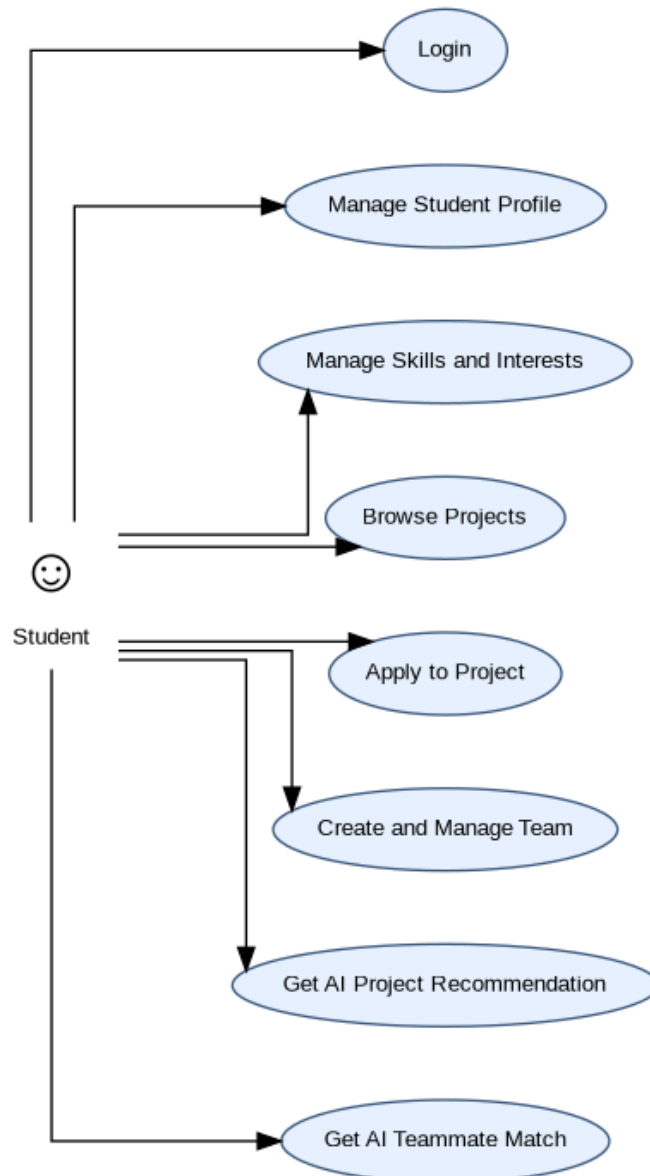
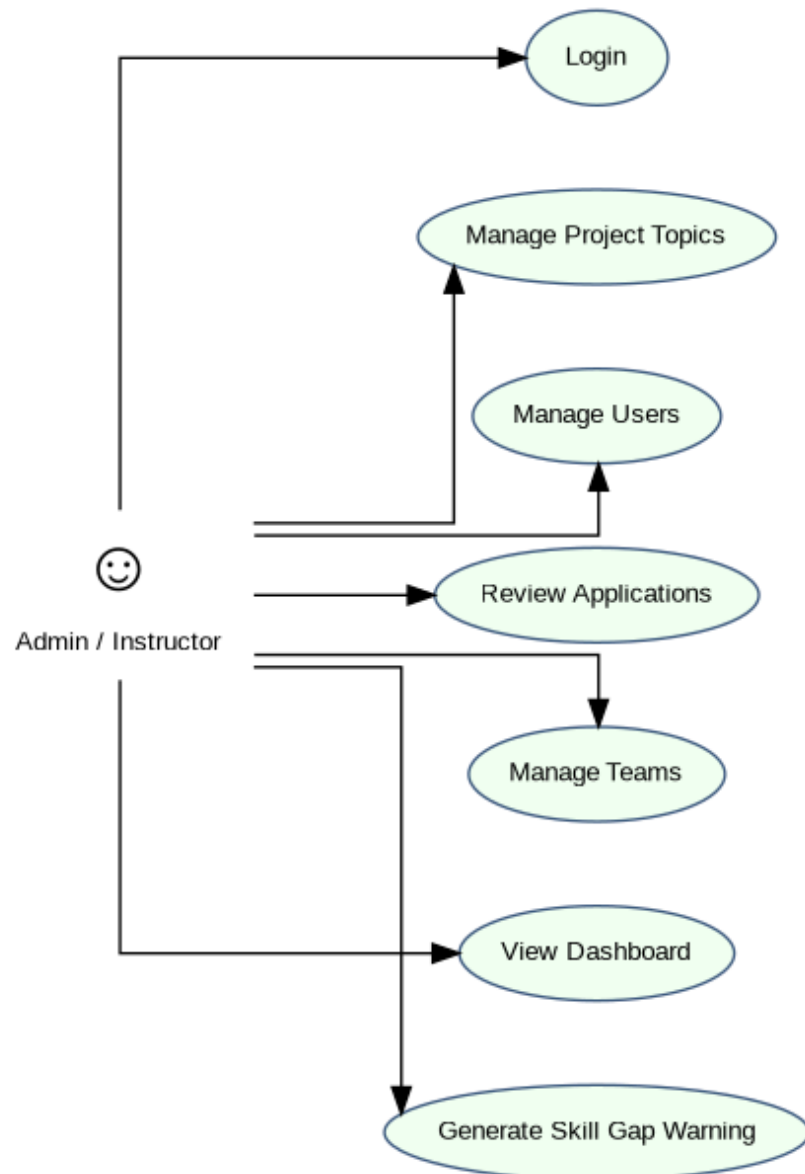


Figure 3. Admin use-case diagram.



4.3 Use-Case Summary

ID	Use Case	Actor(s)	Brief Description	Priority
UC-01	Login	Student, Admin	Allows a registered user to access protected system functions.	High
UC-02	Manage Student Profile	Student	Allows a student to create, view, update, and maintain profile information.	High
UC-03	Manage Skills and Interests	Student	Allows a student to add, update, and remove skills, interests, and preferred technologies.	High
UC-04	Manage Project Topics	Admin	Allows an admin to create, view, update, and delete project topics.	High
UC-05	Browse and Search Projects	Student	Allows students to view and filter available project topics.	High
UC-06	Apply to Project	Student	Allows students to submit project applications.	High
UC-07	Create and Manage Team	Student, Admin	Allows students or admins to create teams and manage team members.	Medium
UC-08	Request AI Project Recommendation	Student, AI Service	Returns ranked project recommendations with explanations.	Medium
UC-09	Request AI Teammate Match	Student, AI Service	Returns possible teammates based on complementary skills and interests.	Medium
UC-10	Review Applications	Admin	Allows an admin to review, approve, reject, or waitlist applications.	High
UC-11	Generate Team Skill Gap Warning	Admin, AI Service	Identifies missing or weak skills in a team compared to project requirements.	Medium
UC-12	View Admin Dashboard	Admin	Shows project, application, team, and student participation status.	Medium

5. Use-Case Descriptions

UC-01 Login

Use Case Number	UC-01
Name	Login
Brief Description	This use case allows a registered student or admin to log in to the system.
Actors	Student, Admin

Preconditions

- The user has a registered account.
- The user is not already logged in.

Basic Flow

Step	User Action	System Response
1	The user opens the login page.	The system displays fields for email and password.
2	The user enters email and password and requests login.	The system validates the credentials.
3	The credentials are valid.	The system starts a session and redirects the user to the correct dashboard based on role.

Postconditions

- The user is authenticated and can access protected features.

Business Rules

- Only registered users may log in.
- The system must not display stored passwords.
- Users must be assigned a role before accessing role based features.

Alternate and Exceptional Flows

- Invalid credentials: The system displays an error message and does not start a session.
- Inactive account: The system blocks login and displays a status message.
- System error: The system displays a generic failure message and logs the error.

Special Requirements

- Password data must be stored securely.

UC-02 Manage Student Profile

Use Case Number	UC-02
Name	Manage Student Profile
Brief Description	This use case allows a student to create, view, and update profile information used for project and teammate matching.
Actors	Student

Preconditions

- The student is logged in to the system.

Basic Flow

Step	User Action	System Response
1	The student selects the profile option.	The system displays the current profile or an empty profile form.
2	The student enters or updates name, program, experience level, project interests, preferred technologies, availability, and short bio.	The system validates required fields and field formats.
3	The student requests to save the profile.	The system checks that the student profile is linked to the logged in user and saves the profile.
4	The profile is saved.	The system displays the updated profile and a success message.

Postconditions

- The student profile is persisted or updated in the system.

Business Rules

- A student can manage only their own profile.
- Each student account can have only one active profile.
- Profile data must be available for recommendation use only after the student saves it.

Alternate and Exceptional Flows

- Missing required fields: The system identifies the missing fields and does not save.
- Duplicate profile: The system blocks creation and asks the student to update the existing profile.
- Database error: The system displays an error message and preserves entered values when possible.

UC-03 Manage Skills and Interests

Use Case Number	UC-03
Name	Manage Skills and Interests
Brief Description	This use case allows a student to add, edit, or remove skills and interests used for recommendation matching.
Actors	Student

Preconditions

- The student is logged in.
- The student profile exists.

Basic Flow

Step	User Action	System Response
1	The student opens the skills and interests page.	The system displays existing skills, proficiency levels, interests, and preferred technologies.
2	The student adds or updates a skill and selects a proficiency level.	The system validates the skill name and level.
3	The student removes an unwanted skill or interest.	The system marks the item for removal.
4	The student saves the changes.	The system updates the student skill and interest records.
5	The update succeeds.	The system displays the current skill profile and a success message.

Postconditions

- The student skills and interests are persisted in the system.

Business Rules

- A student cannot create duplicate skills in the same profile.
- Skill proficiency must be selected from approved values such as Beginner, Intermediate, and Advanced.
- Skills may be used by the AI recommendation service.

Alternate and Exceptional Flows

- Duplicate skill: The system displays a duplicate skill message.
- Invalid proficiency: The system asks the student to select an approved value.
- No profile exists: The system redirects the student to create a profile first.

UC-04 Manage Project Topics

Use Case Number	UC-04
Name	Manage Project Topics
Brief Description	This use case allows an admin to create, view, update, and delete project topics.
Actors	Admin / Instructor

Preconditions

- The admin is logged in.
- The admin has permission to manage project topics.

Basic Flow

Step	User Action	System Response
1	The admin opens project topic management.	The system displays a list of project topics and commands to create or edit topics.
2	The admin requests to create a topic.	The system displays a project topic form.
3	The admin enters title, description, required skills, difficulty, maximum team size, and status.	The system validates required fields and business rules.
4	The admin requests to save the project topic.	The system checks for duplicate active topic titles and saves the topic.
5	The topic is saved.	The system displays the updated project topic list.

Postconditions

- The project topic is created or updated in the system.

Business Rules

- Only admins can create, update, or delete project topics.
- An active project topic title must be unique within the course.
- A project topic cannot be deleted after approved applications exist unless the admin first cancels or reassigns those applications.

Alternate and Exceptional Flows

- Duplicate title: The system displays a duplicate title message.
- Missing required skills: The system asks the admin to provide at least one required skill.

- Topic has applications: The system blocks deletion and displays the reason.

UC-05 Browse and Search Projects

Use Case Number	UC-05
Name	Browse and Search Projects
Brief Description	This use case allows a student to view and filter available project topics.
Actors	Student

Preconditions

- The student is logged in.
- At least one project topic exists.

Basic Flow

Step	User Action	System Response
1	The student opens the project catalog.	The system displays available project topics.
2	The student enters search text or selects filters such as skill, difficulty, category, or team size.	The system filters the project list.
3	The student selects a project topic.	The system displays project details, required skills, team size, and application status.
4	The student returns to the list.	The system preserves or clears filters based on the student action.

Postconditions

- The student can view project details and decide whether to apply.

Business Rules

- Only active project topics are visible to students by default.
- Closed project topics cannot accept new applications.
- Search results must not expose hidden admin notes.

Alternate and Exceptional Flows

- No results found: The system displays a no results message.
- Project closed: The system displays the topic but disables application if allowed by admin settings.

UC-06 Apply to Project

Use Case Number	UC-06
Name	Apply to Project
Brief Description	This use case allows a student to submit an application for a project topic.
Actors	Student

Preconditions

- The student is logged in.
- The student profile exists.
- The selected project topic is open for applications.

Basic Flow

Step	User Action	System Response
1	The student views a project topic.	The system displays the application option if the topic is open.
2	The student selects apply.	The system displays an application form with optional note or motivation.
3	The student enters the application note and submits.	The system verifies that the student has not already applied to the same project.
4	The application is valid.	The system saves the application with Pending status.
5	The application is saved.	The system displays the application status to the student.

Postconditions

- A project application exists in Pending status.

Business Rules

- A student may have only one active application for the same project.
- Applications can be submitted only while the project is open.
- A student cannot apply without a saved profile.

Alternate and Exceptional Flows

- Duplicate application: The system displays the existing application status.
- Missing profile: The system asks the student to create a profile first.
- Project closed: The system blocks submission and displays a closed project message.

UC-07 Create and Manage Team

Use Case Number	UC-07
Name	Create and Manage Team
Brief Description	This use case allows students and admins to create teams and manage team membership.
Actors	Student, Admin / Instructor

Preconditions

- The user is logged in.
- The project topic exists.
- Team formation is open.

Basic Flow

Step	User Action	System Response
1	The student or admin opens the team management page.	The system displays existing teams and available actions.
2	The user creates a team for a project or selects an existing team.	The system displays team details and current members.
3	The user adds or invites students to the team.	The system validates team size and membership rules.
4	The user saves the team changes.	The system updates team membership and status.
5	The update succeeds.	The system displays the updated team list.

Postconditions

- A team is created or team membership is updated.

Business Rules

- A student can belong to only one active team for the same course project cycle.
- A team cannot exceed the maximum team size set for the project.
- Admins can override team assignment when required.

Alternate and Exceptional Flows

- Team full: The system blocks adding another member.
- Student already assigned: The system displays the existing team assignment.
- Project not open for teams: The system blocks team creation.

UC-08 Request AI Project Recommendation

Use Case Number	UC-08
Name	Request AI Project Recommendation
Brief Description	This use case allows a student to request ranked project recommendations with explanations.
Actors	Student, AI Recommendation Service

Preconditions

- The student is logged in.
- The student profile has saved skills and interests.
- At least one active project topic exists.

Basic Flow

Step	User Action	System Response
1	The student selects AI project recommendations.	The system retrieves the student profile, skills, interests, and active project topics.
2	The system sends the relevant matching data to the AI recommendation service.	The AI service ranks project topics and returns short explanations.
3	The system receives AI recommendation results.	The system stores the recommendation result and displays ranked projects.
4	The student views a recommended project.	The system displays the project details and explanation.

Postconditions

- The student sees ranked project recommendations and explanations.

Business Rules

- AI recommendations are advisory only.
- The system must not automatically approve applications based on AI output.
- Only data needed for matching should be sent to the AI service.

Alternate and Exceptional Flows

- Insufficient profile data: The system asks the student to complete skills and interests.
- AI service unavailable: The system displays an unavailable message and allows normal project browsing.
- No active projects: The system displays a no project available message.

Special Requirements

- AI recommendation results should be explainable in short plain language.

UC-09 Request AI Teammate Match

Use Case Number	UC-09
Name	Request AI Teammate Match
Brief Description	This use case allows a student to request suggested teammates based on shared interests and complementary skills.
Actors	Student, AI Recommendation Service

Preconditions

- The student is logged in.
- The student profile exists.
- Other students have profiles available for matching.

Basic Flow

Step	User Action	System Response
1	The student selects teammate matching.	The system retrieves the student profile and available student profiles.
2	The system sends matching data to the AI recommendation service.	The AI service identifies possible teammate matches.
3	The system receives teammate suggestions.	The system displays ranked teammate suggestions and reasons.
4	The student views a teammate suggestion.	The system displays public profile information and matching explanation.

Postconditions

- The student sees possible teammates and match explanations.

Business Rules

- Private data must not be exposed in teammate suggestions.
- A student already assigned to a full team should not appear as available unless admin settings allow it.
- AI suggestions are advisory only.

Alternate and Exceptional Flows

- No available teammates: The system displays a no matches message.
- AI service unavailable: The system displays an error and allows manual browsing.
- Incomplete profile: The system asks the student to update profile data.

UC-10 Review Applications

Use Case Number	UC-10
Name	Review Applications
Brief Description	This use case allows an admin to review student project applications and update application status.
Actors	Admin / Instructor

Preconditions

- The admin is logged in.
- At least one project application exists.

Basic Flow

Step	User Action	System Response
1	The admin opens the application review page.	The system displays applications grouped by project topic or status.
2	The admin selects an application.	The system displays student profile summary, project details, and application note.
3	The admin chooses approve, reject, or waitlist.	The system validates the project capacity and application status.
4	The admin submits the decision.	The system updates the application status and records the decision.
5	The update succeeds.	The system displays the updated application list.

Postconditions

- The selected application status is updated.

Business Rules

- Only admins can approve or reject applications.
- Approved applications must not exceed project or team capacity unless overridden by admin.
- Application decisions should be recorded for audit and review.

Alternate and Exceptional Flows

- Project capacity full: The system blocks approval unless admin override is allowed.
- Application already decided: The system displays the current status.
- Student already assigned: The system displays the existing assignment.

6. Supplementary Specifications

6.1 Functionality Requirements

- The system shall provide role based access for Student and Admin users.
- The system shall provide CRUD operations for student profiles, skills, project topics, applications, and teams.
- The system shall keep AI recommendation history for review and debugging when enabled.

6.2 Usability Requirements

- Students shall be able to create a profile and apply to a project with minimal steps.

- Admins shall be able to create a project topic without technical knowledge.
- Validation messages shall clearly explain what needs to be corrected.

6.3 Reliability Requirements

- The main system shall remain usable even when the AI service is unavailable.
- The system shall prevent duplicate records where business rules require uniqueness.
- The system shall preserve submitted data when validation fails where practical.

6.4 Performance Requirements

- Normal project search and listing pages should respond quickly for a course sized dataset.
- AI recommendations may take longer than normal operations but must return a result or clear error message.
- The system should support multiple students using the application during the same project selection period.

6.5 Security Requirements

- Only authenticated users shall access protected pages.
- Students shall update only their own profiles and applications.
- Admins shall manage project topics, applications, users, and teams.
- Passwords shall not be stored as plain text.
- Only necessary matching data shall be shared with the AI service.

6.6 Supportability Requirements

- The project shall include a README with setup instructions.
- The project shall include database setup instructions.
- The project shall include basic test cases for main use cases.
- The project artifacts shall be committed and pushed to GitHub regularly.

6.7 Design Constraints

- The system shall be web based.
- The backend should use a course approved technology stack.
- The AI feature may use an external AI API or local AI model.
- The first release targets one course or project cycle.

7. Glossary

Term	Definition
Admin / Instructor	User who manages project topics, student applications, teams, and dashboards.
AI Recommendation Service	External or internal service that generates project matches, teammate matches, and skill gap warnings.
Application	A student request to join or work on a specific project topic.
Project Topic	A course project idea created by an admin with title, description, difficulty, and required skills.
Recommendation	A ranked suggestion generated by the system or AI service.
Skill Gap Warning	A warning that a team may be missing skills required for its selected project.
Student Profile	Student information used for project and teammate matching.
Team	A group of students assigned to work on a selected project topic.
Use Case	A goal oriented interaction between an actor and the system.

8. Use Case to Feature Traceability

Feature	Related Use Case(s)
Student profile management	UC-02, UC-03
Project topic management	UC-04, UC-05
Project application tracking	UC-06, UC-10
Team formation	UC-07, UC-09, UC-11

Feature	Related Use Case(s)
AI based project recommendation	UC-08
AI based teammate matching	UC-09
Admin monitoring	UC-10, UC-11, UC-12
Authentication and role access	UC-01